



# Introduction to Standard Deviation

Consider the following numbers:

7 7 6 8 7

How do we calculate the average (mean)?

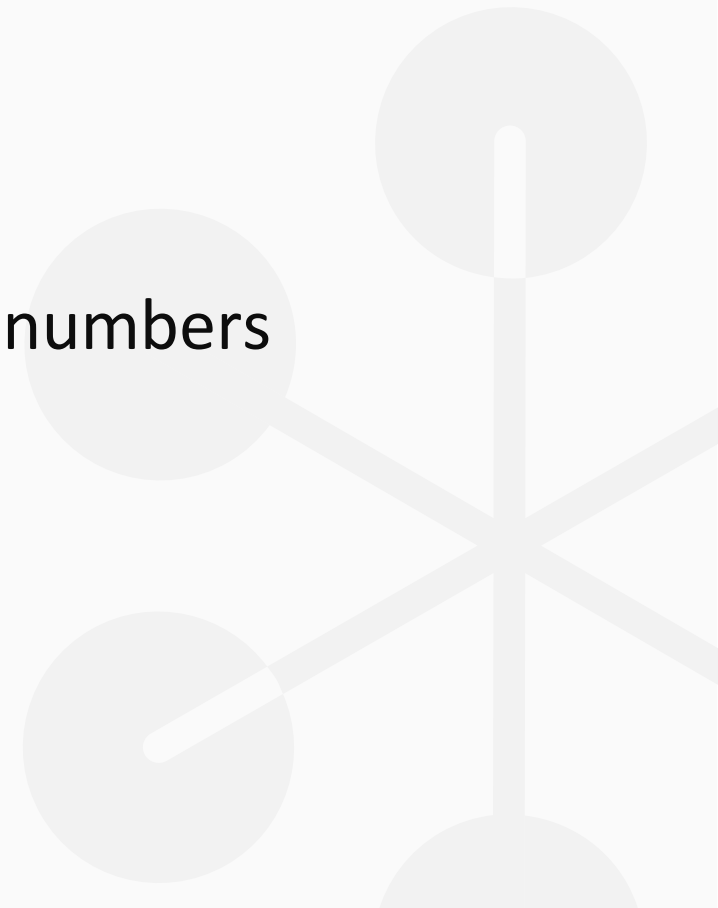


Consider the following numbers:

7 7 6 8 7

How do we calculate the average (mean)?

We add the numbers together, then divide by how many numbers there are.



Find the mean

7 7 6 8 7

$$7 + 7 + 6 + 8 + 7 = 35$$

$$35 \div 5 = 7$$

So, the mean is 7.

Let's try it for another set of numbers.

Find the mean for another set of numbers

9 6 9 3 8

$$9 + 6 + 9 + 3 + 8 = 35$$

$$35 \div 5 = 7$$

So, the mean for this set of numbers is also 7.



Same mean

7 7 6 8 7  
9 6 9 3 8

The mean for both sets is 7, despite them having an obviously different spread.

The first one is like a wiggle, the second is like a stagger!.

# Introducing variance

- Variance is a measure of this spread of values.
- If you did not enjoy mathematics at school, you might find the equation for calculating variance intimidating, but we will break it down.

$$\textit{variance} = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$$

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$\bar{x}$  is the mean value – in the two examples,  $\bar{x}$  is 7

$n$  is the count of numbers – 5 in the two examples.

$x_i$  represents each of the numbers in turn.

$i$  is the index.



$$\text{variance} = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$$

- $x_i$  represents each number in turn.  $i$  is the index.
- In the second example, 9 6 9 3 8
- $x_1$  is 9
- $x_2$  is 6
- $x_3$  is 9
- $x_4$  is 3
- $x_5$  is 8

$$\textit{variance} = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$$

- The Greek letter  $\Sigma$  (capital sigma) is an instruction to find the sum of everything to the right.
- The lower  $i=1$  tells us to begin with the index 1, and the upper  $n$  tells us to end with the index  $n$  (5 in this example).
- So, for each number in turn, we subtract the mean, square the result, and keep a running total.
- Note that a negative number squared always gives a positive number.
- Once completed, we divide by  $n$ . This gives us the variance.

Let's see this in action using Excel:

Using Excel for 7, 7, 6, 8, 7

|   | A         | B             | C    | D        | E       |
|---|-----------|---------------|------|----------|---------|
| 1 | Index (i) | No. ( $x_i$ ) | Mean | No.-Mean | squared |
| 2 | 1         | 7             | 7    | 0        | 0       |
| 3 | 2         | 7             | 7    | 0        | 0       |
| 4 | 3         | 6             | 7    | -1       | 1       |
| 5 | 4         | 8             | 7    | 1        | 1       |
| 6 | 5         | 7             | 7    | 0        | 0       |
| 7 |           |               |      |          |         |
| 8 |           |               |      | Sum:     | 2       |
| 9 |           |               |      | Sum ÷ n: | 0.4     |

Using Excel for 9, 6, 9, 3, 8

|   | A         | B             | C    | D        | E       |
|---|-----------|---------------|------|----------|---------|
| 1 | Index (i) | No. ( $x_i$ ) | Mean | No.-Mean | squared |
| 2 | 1         | 9             | 7    | 2        | 4       |
| 3 | 2         | 6             | 7    | -1       | 1       |
| 4 | 3         | 9             | 7    | 2        | 4       |
| 5 | 4         | 3             | 7    | -4       | 16      |
| 6 | 5         | 8             | 7    | 1        | 1       |
| 7 |           |               |      |          |         |
| 8 |           |               |      | Sum:     | 26      |
| 9 |           |               |      | Sum ÷ n: | 5.2     |

# Variance and Standard Deviation

- Standard Deviation is the square root of Variance.
- Once we know the variance, we can find the square root of it using a calculator or Excel.

# Variance and Standard Deviation

- Taking the square root of the variance, we find the standard deviation.
- In the first example (7, 7, 6, 8, 7), the variance is 0.4, so the standard deviation is 0.63
- In the second example (9, 6, 9, 3, 8), the variance is 5.2, so the standard deviation is 2.28
- The larger number is in keeping with the bigger wiggle!

# Population and Sample

- In the above examples, the calculations are carried out on all of the numbers.
- This is Population Standard Deviation.
- It is more usual to obtain data from a sample, rather than the entire population.
- For example, there are approximately 70 million people in the UK.

# Population and Sample

- If you wanted to know the mean and standard deviation of people's
  - height
  - weight
  - number of hours spent using their phone
  - anything else
- it would probably not be practical to survey all 70 million!
- Instead, a sample of the population would be surveyed, resulting in an approximate variance and standard deviation.

# Sample Variance and Standard Deviation

$$\textit{variance} = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$$

- The formula for calculating variance for a sample is similar to the formula for calculating variance for a population.
- Note that the top is divided by one less than the total number of items in the sample.
- As before, the Standard Deviation is the square root of the Variance.

## Example of a sample calculation

- An email marketing team tracked the conversion rates for a specific campaign over eight consecutive weeks.
- We can assume that the campaign was over a much longer period, hence the eight weeks provide a sample.

| Week | Conversion Rate (%) |
|------|---------------------|
| 1    | 4.2                 |
| 2    | 5.5                 |
| 3    | 4.8                 |
| 4    | 5.1                 |
| 5    | 6.0                 |
| 6    | 4.5                 |
| 7    | 5.3                 |
| 8    | 5.8                 |



# Calculate the mean

- As before, we calculate the mean by adding the data points and dividing by the number of data points.
- Sum of rates:  $4.2 + 5.5 + 4.8 + 5.1 + 6.0 + 4.5 + 5.3 + 5.8 = 41.2$
- Number of data points (n): 8
- Mean ( $\bar{x}$ ):  $41.2/8 = 5.15$



Calculate each point's deviation from the mean

| Week | $x_i$ | $(x_i - \bar{x})$    |
|------|-------|----------------------|
| 1    | 4.2   | $4.2 - 5.15 = -0.95$ |
| 2    | 5.5   | $5.5 - 5.15 = 0.35$  |
| 3    | 4.8   | $4.8 - 5.15 = -0.35$ |
| 4    | 5.1   | $5.1 - 5.15 = -0.05$ |
| 5    | 6.0   | $6.0 - 5.15 = 0.85$  |
| 6    | 4.5   | $4.5 - 5.15 = -0.65$ |
| 7    | 5.3   | $5.3 - 5.15 = 0.15$  |
| 8    | 5.8   | $5.8 - 5.15 = 0.65$  |

# Square each deviation

| Week | $(x_i - \bar{x})$ | $(x_i - \bar{x})^2$  |
|------|-------------------|----------------------|
| 1    | -0.95             | $(-0.95)^2 = 0.9025$ |
| 2    | 0.35              | $(0.35)^2 = 0.1225$  |
| 3    | -0.35             | $(-0.35)^2 = 0.1225$ |
| 4    | -0.05             | $(-0.05)^2 = 0.0025$ |
| 5    | 0.85              | $(0.85)^2 = 0.7225$  |
| 6    | -0.65             | $(-0.65)^2 = 0.4225$ |
| 7    | 0.15              | $(0.15)^2 = 0.0225$  |
| 8    | 0.65              | $(0.65)^2 = 0.4225$  |

## Sum the squared deviations

- $0.9025 +$
- $0.1225 +$
- $0.1225 +$
- $0.0025 +$
- $0.7225 +$
- $0.4225 +$
- $0.0225 +$
- $0.4225 = \underline{2.74}$



Divide the sum by  $n - 1$

- This gives us the Sample Variance
- $2.74/7 = 0.3914$

## Take the square root of the Sample Variance

- This gives us the Sample Standard Deviation
- $\sqrt{0.3914} = 0.6256$ , or approximately 0.63%



# Interpretation

- The sample standard deviation for the email campaign's conversion rates is approximately **0.63%**.
- This means that, on average, the weekly conversion rates varied by about 0.63 percentage points from the mean of 5.15%.
- A smaller standard deviation indicates that the campaign's performance was relatively consistent over the measured period.

# Standard Deviation and Confidence Intervals

- The diagram shows the Normal Distribution Curve.
- The marks along the X-Axis represent Standard Deviations from the mean value (given here as the Greek letter  $\mu$ ).



# Standard Deviation and Confidence Intervals

- Confidence Intervals can be used to determine how likely a value or parameter (the mean) of a smaller sample is representative of a population.



# Standard Deviation and Confidence Intervals

- As the diagram shows, 2 SDs (i.e. 1 above and 1 below the mean) gives us a confidence of 68%.
- 4 SDs give 95% confidence.
- 6 SDs give 99.7% confidence.



# Standard Deviation and Confidence Intervals

- The marks along the X-Axis represent Standard Deviations from the mean value (given here as the Greek letter  $\mu$ ).

